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Finite-time adaptive sliding mode tracking control for a quadrotor unmanned aerial vehicle with model uncertainties and external disturbances

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Abstract A finite-time adaptive sliding mode controller is designed to improve the trajectory tracking capability of a quadrotor unmanned aerial vehicle (UAV) in the presence of model uncertainties and unmodeled external disturbances. By applying adaptive nonsingular fast terminal sliding mode control (ANFTSMC) to the quadrotor UAV system, rapid response, finite-time convergence, and high-precision steady-state control are achieved under uncertainties and external disturbances, while also avoiding singularities and reducing the effect of chattering. Furthermore, the adaptive strategy is designed to estimate the upper bounds of model uncertainties and external disturbances without requiring prior bounded information. The finite-time convergence capability of the system is rigorously analyzed using the Lyapunov stability framework. Extensive simulation studies and experimental validations collectively demonstrate the efficacy of the designed control scheme. Notably, stable flight in the presence of wind disturbances and control of the contact force in experiments with wind turbine blades underscore the applicability of the strategy for achieving reliable quadrotor UAV operations in complex environments.

Keywords quadrotor, UAV, uncertainties and disturbances, ANFTSMC

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1 Introduction

Unmanned aerial vehicles (UAVs) are increasingly deployed in both civilian and military applications, owing to their adaptability, low operational costs, and proficiency in conducting missions in high-risk environments where human presence is either unviable or poses significant dangers. Quadrotors have emerged as the most popular choice among various types of UAVs, given their simple mechanical structure, high maneuverability, and vertical takeoff and landing capabilities, which make them ideal for a wide range of applications, such as aerial surveillance [1], environmental monitoring [2], remote sensing data collection [3], and structure inspection [4]. Nevertheless, quadrotor UAVs are complex systems because of their underactuated design and nonlinear, coupled dynamics. Moreover, environmental disturbances such as atmospheric turbulence and wind gusts, combined with dynamic parameter uncertainties, may lead to flight trajectory deviation or system instability [5,6].

Significant research efforts have been devoted to addressing the challenges associated with trajectory tracking in quadrotor UAVs. Despite the efficacy of traditional linear control strategies, including proportional-integral-derivative (PID) and linear quadratic regulator (LQR) techniques, for

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systems with linear dynamics, maintaining stability becomes problematic for quadrotors operating outside their designated domains, especially when tracking nonlinear target trajectories [7]. Some nonlinear control methodologies have demonstrated efficacy in quadrotor applications, including backstepping-based approaches [8] and fuzzy logic control (FLC) [9]. However, these approaches often rely on accurate system models and lack adaptability in dynamic environments [10]. Controlling quadrotor UAVs under external disturbances is further complicated by challenges such as in-flight vibrations, strongly coupled nonlinear system dynamics, and unexpected external disturbances (e.g., wind gusts, environmental contact forces, or sudden load changes). Additionally, UAVs must be able to operate in real-time with rapid response capabilities [11]. Sliding mode control (SMC) is distinguished by its inherent disturbance rejection capabilities against nonlinear structures, unmodeled dynamics, and external perturbations, making it fairly effective in maintaining the stability of quadrotor UAVs across various operating conditions. The relatively straightforward design of SMC further facilitates its implementation in complex UAV dynamics [12,13].

To enhance the disturbance rejection capability of the system, a sufficiently large fixed switching gain has been employed in traditional SMC to account for the maximum possible magnitude of model uncertainties and external perturbations [14]. However, in real-world implementations, the bounds of uncertainties and disturbances cannot be predetermined [15,16]. Moreover, overestimating the control gain may lead to chattering phenomena, thereby adversely affecting the system performance [17]. Adaptive SMC (ASMC) has emerged as an alternative strategy for dynamically tuning the control gain based on real-time estimation of uncertainties and external disturbances. Although ASMC designs reported in refs. [18,19] reduced chattering without requiring disturbance bounds, they still exhibited a slow perturbation response, limiting the convergence rate and steady-state accuracy.

To improve the convergence rate and realize finite-time convergence, which are critical requirements for time-sensitive quadrotor tasks, researchers developed terminal SMC (TSMC) by incorporating a terminal attractor-based approach [20]. Recent advances have integrated TSMC with complementary techniques to address specific challenges. For instance, Najafi et al. [21] and Alattas et al. [22] combined barrier functions with finite-time SMC to handle state constraints and actuator faults, whereas the convergence speed was regulated using quantized predefined-time control [23]. However, these methods often produce conservative or aggressive control inputs, compromising the transient performance, especially near constraint boundaries or predefined settling times. Neural network-based approaches [24–26] have been developed to tackle complex uncertainties, but their inherent approximation lag hinders timely compensation for fast-varying disturbances (e.g., sudden wind gusts), which are commonly encountered during quadrotor operations. A prior study [27] demonstrated that TSMC coupled with adaptive mechanisms enables stable attitude control despite actuator saturation effects. Nevertheless, the performance of TSMC is degraded in the event of large deviations from equilibrium. To address this issue, fast terminal SMC (FTSMC) was developed, which accelerates the error convergence by employing supplementary sliding surface terms. Despite this improvement, FTSMC faces singularity issues when states approach the origin. This has been resolved by optimizing the design of the sliding surface using nonsingular FTSMC (NFTSMC) [28]. Further, a previous study [29] introduced an adaptive NFTSMC (ANFTSMC) for robust quadrotor tracking. However, the adaptive parameters failed to decrease with diminishing disturbances, leading to overestimation [30].

Based on the aforementioned discussions, in this study, an ANFTSMC framework is developed for enhancing the trajectory tracking of quadrotor UAVs in the event of parametric uncertainties and environmental disturbances. As the foundational work, a complete dynamical model accounting for both system modeling uncertainties and environmental interference factors is established. The detailed design of the proposed ANFTSMC is presented, in which a nonsingular fast terminal sliding surface is integrated with adaptive laws to compensate for external disturbances and system uncertainties. It is demonstrated that the adaptive laws enable the sliding mode variable to converge to and remain within a bounded region of the sliding manifold in finite time. The proposed controller is compared with existing approaches via simulation studies, demonstrating both its theoretical validity and performance advantages. Finally, its favorable tracking performance is further demonstrated through comparative experiments under wind disturbances. The system is also successfully applied in controlling the contact force on wind turbine blades. The primary contributions of this study are as follows.

(1) A finite-time adaptive control approach is proposed to achieve rapid response, finite-time convergence, and high-precision steady-state control of a quadrotor UAV with model uncertainties and external disturbances.

(2) An adaptive strategy is designed to estimate the bounds of model uncertainties and external disturbances, without requiring prior knowledge of the bounds.

(3) The effectiveness of the proposed control scheme is validated in real-world test flights of the quadrotor under wind disturbances and is further verified in scenarios involving physical interaction and regulation of the contact force on a wind turbine blade.

This paper is organized as follows: Section 2 presents the dynamic model of the quadrotor UAV under the influence of

state-dependent uncertainties and external disturbances. Section 3 details the design and stability analysis of the adaptive NFTSMC controller. Section 4 presents a simulation comparison of various controllers. Section 5 presents experimental comparisons of the controllers under disturbance, along with verification of the effectiveness of the proposed controller in force control applications. Finally, Section 6 concludes the study.

2 System dynamics

2.1 Dynamic modeling of quadrotor UAV

The coordinate frames of the quadrotor UAV were established as shown in Figure 1. $O_I\{x_i, y_i, z_i\}$ represents the inertial coordinate frame; $O_b\{x_b, y_b, z_b\}$ and $O_e\{x_e, y_e, z_e\}$ represent the quadrotor body-fixed coordinate frame and the end effector coordinate frame, respectively. The generalized coordinate variables consist of position coordinate variables and attitude coordinate variables. Let $\mathbf{p} = [x, y, z]^{\mathrm{T}}$ denote the position of the quadrotor, and let the Euler angles $\mathbf{\Phi} = [\phi, \theta, \psi]^{\mathrm{T}}$ denote the attitude of the quadrotor, both of which are represented in the inertial coordinate frame O_I. Thus, the generalized variables can be expressed as follows:

$$\mathbf{q} = [\mathbf{p}^{\mathrm{T}}, \mathbf{\Phi}^{\mathrm{T}}]^{\mathrm{T}}. \tag{1}$$

Based on the established coordinate system, the dynamic model of the quadrotor in the presence of external disturbances can be expressed as follows [31]:

$$\mathbf{H}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{u} + \mathbf{D}_{\mathrm{ext}}, \tag{2}$$

where $\mathbf{H}(\mathbf{q}) \in \mathbb{R}^{6\times 6}$ is a positive definite symmetric inertia matrix, and its structure is determined by the mass and moments of inertia of the quadrotor UAV. $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{6\times 6}$ is the Coriolis and centripetal matrix. This matrix captures the gyroscopic effects resulting from rotor rotation and the coupling terms between the translational and rotational dynamics. $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^6$ is the gravitational vector representing the effect of gravitational forces on the generalized coordinates of the system. $\mathbf{u} \in \mathbb{R}^6$ denotes the generalized input force and torque. $\mathbf{D}_{\mathrm{ext}} \in \mathbb{R}^6$ represents the external environmental disturbances acting on the system. Beyond such external disturbances, the system itself inherently suffers from modeling uncertainties, which can be expressed as

$$\begin{cases} \mathbf{H}(\mathbf{q}) = \mathbf{H}_0(\mathbf{q}) + \Delta\mathbf{H}(\mathbf{q}), \\ \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}}) + \Delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}), \\ \mathbf{G}(\mathbf{q}) = \mathbf{G}_0(\mathbf{q}) + \Delta\mathbf{G}(\mathbf{q}), \end{cases} \tag{3}$$

where $\mathbf{H}_0(\mathbf{q})$, $\mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})$, and $\mathbf{G}_0(\mathbf{q})$ represent the nominal components of $\mathbf{H}(\mathbf{q})$, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, and $\mathbf{G}(\mathbf{q})$, respectively; and $\Delta\mathbf{H}_0(\mathbf{q})$, $\Delta\mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})$, and $\Delta\mathbf{G}_0(\mathbf{q})$ stand for the uncertain components of the corresponding model matrices/vectors, which originate from factors including unmodeled dynamic characteristics, parameter fluctuations, and measurement errors in the system. Thus, these components quantify the deviation between the actual system model and the nominal model.

By substituting the results of eq. (3) into the dynamic model eq. (2), the following formula can be obtained:

$$\mathbf{H}_0(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}_0(\mathbf{q}) = \mathbf{u} + \mathbf{D}_{\mathrm{lum}}, \tag{4}$$

where $\mathbf{D}_{\mathrm{lum}} = \mathbf{D}_{\mathrm{ext}} - \mathbf{H}_0(\mathbf{q})\ddot{\mathbf{q}} - \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}_0(\mathbf{q})$ denotes the lumped disturbances, which include disturbance to the system caused by the external environment and the modeling uncertainty generated when determining the nominal physical parameters of $\mathbf{H}(\mathbf{q})$, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, $\mathbf{G}(\mathbf{q})$. The input forces and torques related to the position $\mathbf{p}$ and attitude $\mathbf{\Phi}$ of the quadrotor, represented by the generalized variables $\mathbf{q}$, must be transformed into actual actuator commands. The position and attitude of the quadrotor are controlled by four motors, each regulating the rotation speed of a propeller. These motors require corresponding four-way pulse width modulation (PWM) signals, which are combined to produce the thrust and torques needed for control.

The input $\mathbf{u}$ comprises two components: $\mathbf{F} = [F_x, F_y, F_z]^{\mathrm{T}}$, representing the total thrust generated by the four propellers in the inertial frame O_I, and $\mathbf{M} = [M_x, M_y, M_z]^{\mathrm{T}}$, representing the torques produced by the propellers in the inertial frame O_I. Thus,

$$\mathbf{u} = \begin{bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} = \underbrace{\begin{bmatrix} {}^I R_b & 0 \\ 0 & Q^{\mathrm{T}} \end{bmatrix}}_{R_F} \begin{bmatrix} 0 \\ 0 \\ {}^b T \\ {}^b M_x \\ {}^b M_y \\ {}^b M_z \end{bmatrix}, \tag{5}$$

where $Q = ({}^I R_b)^{\mathrm{T}} R_{\Phi}$; ${}^I R_b$ is the rotation matrix from frame

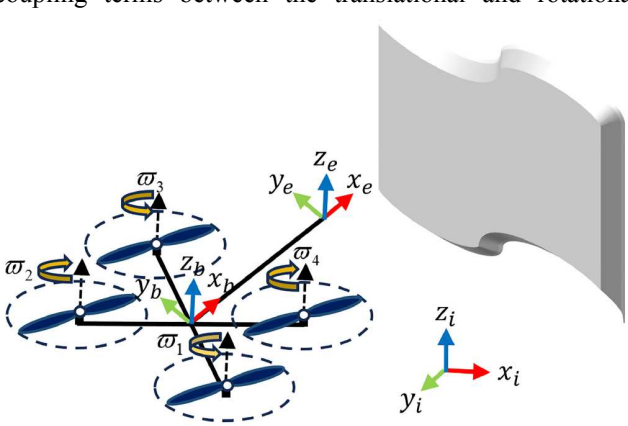

Figure 1 (Color online) Definitions of the coordinate frames for the quadrotor UAV.

O_I to frame O_b, and $R_\Phi = \begin{bmatrix} \sin\theta & 0 & -\cos\phi\sin\theta \\ 0 & 1 & \sin\phi \\ \sin\theta & 0 & \cos\phi\cos\theta \end{bmatrix}$ is a transformation matrix that transforms the Euler angle rates $\dot{\mathbf{\Phi}}$ into the angular velocity $\boldsymbol{\omega}$ in the frame O_I; ${}^bM = \left[{}^bM_x, {}^bM_y, {}^bM_z\right]^{\mathrm{T}}$ represents the total thrust and torques generated by the four propellers, expressed in the body frame O_b. The transformation relationship between the thrust force and torque produced by the propellers and their rotational speed can be described by

$$\begin{bmatrix} 0 \\ 0 \\ {}^bT \\ {}^bM_x \\ {}^bM_y \\ {}^bM_z \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ C_T & C_T & C_T & C_T \\ 0 & -dC_T & 0 & dC_T \\ dC_T & 0 & -dC_T & 0 \\ C_M & -C_M & C_M & -C_M \end{bmatrix}}_{N} \begin{bmatrix} \overline{w}_1^2 \\ \overline{w}_2^2 \\ \overline{w}_3^2 \\ \overline{w}_4^2 \end{bmatrix}, \tag{6}$$

where C_T and C_M are the constant coefficients of the thrust force and torque, respectively, $d \in \mathbb{R}^+$ represents the distance from the center of the quadrotor body to any propeller motor, and $\overline{w}_i$ represents the rotational speed of the ith propeller. By combining eqs. (5) and (6), the ideal input command $\boldsymbol{\tau} \in \mathbb{R}^6$ for the motor actuators can be derived as

$$\boldsymbol{\tau} = \begin{bmatrix} \overline{w}_1^2 \\ \overline{w}_2^2 \\ \overline{w}_3^2 \\ \overline{w}_4^2 \end{bmatrix} = (\mathbf{R_F N})^+ \mathbf{u}, \tag{7}$$

where $(\mathbf{R_F N})^+$ denotes the pseudo-inverse matrix of the product of matrices $\mathbf{R_F} \in \mathbb{R}^{6\times 6}$ and $\mathbf{N} \in \mathbb{R}^{6\times 4}$.

The quadrotor is an underactuated system with six outputs, comprising position $\mathbf{p} \in \mathbb{R}^3$ and attitude $\mathbf{\Phi} \in \mathbb{R}^3$, but it has only four independent inputs. Consequently, the quadrotor can track only four desired commands: the desired position $\mathbf{p_d} \in \mathbb{R}^3$ and yaw angle $\psi_d \in \mathbb{R}$. The remaining two desired attitude variables, roll angle $\phi_d \in \mathbb{R}$ and pitch angle $\theta_d \in \mathbb{R}$, are determined by the desired position $\mathbf{p_d}$, yaw angle ψ_d, and the thrust force components F_x, F_y, and F_z. From eq. (5), it can be deduced that F_x, F_y, F_z, and ${}^bT$ must satisfy the relationship: $\left[F_x, F_y, F_z\right]^{\mathrm{T}} = {}^I R_{b,d}[0, 0, {}^bT]^{\mathrm{T}}$. This suggests that the following equation must hold:

$$\begin{cases} F_x = {}^bT(\cos\psi_d\sin\theta_d\cos\phi_d + \sin\psi_d\sin\phi_d), \\ F_y = {}^bT(\sin\psi_d\sin\theta_d\cos\phi_d - \cos\psi_d\sin\phi_d), \\ F_z = {}^bT\cos\phi_d\cos\theta_d, \end{cases} \tag{8}$$

where the subscript d denotes the desired value. Therefore, the desired roll angle ϕ_d and pitch angle θ_d can be calculated as follows:

$$\begin{cases} \phi_d = \arcsin\left(\dfrac{F_x}{{}^bT}\sin\psi_d - \dfrac{F_y}{{}^bT}\cos\psi_d\right), \\ \theta_d = \arctan\left(\dfrac{F_x}{F_z}\cos\psi_d + \dfrac{F_y}{F_z}\sin\psi_d\right). \end{cases} \tag{9}$$

2.2 Dynamic properties and assumptions

In many practical applications of Euler-Lagrange systems, eq. (4) exhibits essential dynamic properties that are instrumental for controller design and stability analysis.

Property 1 [31] The matrix $\mathbf{H_0(q)}$ is symmetric and uniformly positive definite.

Property 2 [32] There exist h_0, c_0, $g_0 \in \mathbb{R}^+$ such that $\|\mathbf{H}_0(\mathbf{q})\| \leqslant h_0$, $\|\mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})\| \leqslant c_0\|\dot{\mathbf{q}}\|$, $\|\mathbf{G}_0(\mathbf{q})\| \leqslant g_0$.

Assumption 1 [33,34] For the uncertain models in eq. (3), it can be assumed that the uncertainty part satisfies the following: there exist h_1, c_1, $g_1 \in \mathbb{R}^+$ such that $\|\Delta\mathbf{H}(\mathbf{q})\| \leqslant h_1$, $\|\Delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\| \leqslant c_1\|\dot{\mathbf{q}}\|$, $\|\Delta\mathbf{G}(\mathbf{q})\| \leqslant g_1$. Furthermore, the external disturbance $\mathbf{D}_{\mathrm{ext}}$ is subject to a bound such that $\|\mathbf{D}_{\mathrm{ext}}\| \leqslant \overline{d}$, where $\overline{d}$ is an unknown constant. According to the dynamic model equation, it is natural to assume that there exist χ_0, χ_1, and $\chi_2 \in \mathbb{R}^+$ such that the generalized input force and torque $\mathbf{u}$ satisfy the formula $\mathbf{u} \leqslant \chi_0 + \chi_1\|\mathbf{q}\| + \chi_2\|\dot{\mathbf{q}}\|^2$.

Remark 1 For the dynamic model of the quadrotor, the nominal values $(\mathbf{H}_0(\mathbf{q}), \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}}), \mathbf{G}_0(\mathbf{q}))$ are obtainable, and thus their associated bounds h_0, c_0, g_0 are also obtainable. However, $\mathbf{D}_{\mathrm{lum}}$ encompasses both external disturbances and model uncertainties, which are bounded in energy yet highly uncertain. This makes obtaining accurate parameters for $\mathbf{D}_{\mathrm{lum}}$ particularly challenging. Thus, the associated coefficients h_1, c_1, g_1, and $\overline{d}$ are considered to exist but remain unknown.

To facilitate the design of a system controller, the following lemmas are given.

Lemma 1 [35] Considering the system described by eq. (2), suppose there exists a continuous function $V(x)$ satisfying $\dot{V}(x) \leqslant -pV^q(x)$, where constants $p > 0$ and $0 < q < 1$, then the system is finite-time stable. If $V(x_0)$ is set as the initial value of $V(x)$, the reaching time can be bounded by $T \leqslant \dfrac{V^{1-q}(x_0)}{p(1-q)}$.

Lemma 2 [36] Considering the system described by eq. (2), suppose there exists a continuous function $V(x)$ satisfying $\dot{V}(x) \leqslant -p_1V^{q_1}(x) + \rho$, where constants $p_1 > 0$,

$0 < q_1 < 1$, and $0 < \rho < \infty$, then the system exhibits finite-time stability. If $V(x_0)$ is set as the initial value of $V(x)$, the reaching time can be bounded by $T \leqslant \frac{V^{1-q_1}(x_0)}{\varepsilon p_1(1-q_1)}$, where $0 < \varepsilon < 1$.

Lemma 3 [36] For any $x_i \in \mathbb{R}$, $i = 1, \cdots, n$ and a positive constant $0 < m < 2$, the following inequality always holds:

$$(x_1^2 + x_2^2 + \cdots + x_n^2)^m \leqslant (|x_1^m| + |x_2^m| + \cdots + |x_n^m|)^2.$$

3 Controller design

In this section, we aim to further improve the convergence performance of the quadrotor to perform tasks in the presence of disturbances and model uncertainties. To achieve this, a robust control method is designed to ensure that the quadrotor follows the predetermined command quickly and accurately. This control strategy adopts a nonsingular fast terminal sliding surface and does not require prior knowledge of bounded uncertainties or rely on precise model parameters.

3.1 Uncertainty analysis for sliding surface

Define $\mathbf{q_d}$ as the target state of the quadrotor, then $\mathbf{e} = \mathbf{q} - \mathbf{q_d}$ is the error of the current state and the target state. The nonsingular fast terminal sliding surface $\mathbf{s}$ is defined as

$$\mathbf{s} = \mathbf{e} + \gamma_1 \mathbf{e}^{[\alpha_1]} + \gamma_2 \dot{\mathbf{e}}^{[\alpha_2]}, \tag{10}$$

where γ_1, γ_2, α_1, and α_2 are positive constants. α_1 and α_2 must be selected to satisfy

$$\begin{cases} \alpha_1 > \alpha_2, \\ 1 < \alpha_2 = \dfrac{l_1}{l_2} < 2, \end{cases} \tag{11}$$

where l_1 and l_2 are positive constants. For a vector $\mathbf{v} \in \mathbb{R}^n$ and a scalar $k > 0$, the notation $\mathbf{v}^{[k]}$ is the abbreviation of the expression:

$$\mathbf{v}^{[k]} = |\mathbf{v}|^k \operatorname{sgn}(\mathbf{v}), \tag{12}$$

where $\mathbf{v}^{[k]}$ and $|\mathbf{v}|^k$ represent $\mathbf{v}^{[k]} := [v_1^{[k]}, \cdots, v_n^{[k]}]^{\mathrm{T}} \in \mathbb{R}^n$ and $|\mathbf{v}|^k := \operatorname{diag}\{|v_1|^k, \cdots, |v_n|^k\} \in \mathbb{R}^{n\times n}$, where v_i indicates the ith element of vector $\mathbf{v}$, and $v_i^{[k]}$ can be regarded as a one-dimensional vector that can be calculated using eq. (12) as follows: $v_i^{[k]} = |v_i|^k \operatorname{sgn}(v_i)$. Whether for a vector or a scalar, it can be verified that when $k > 1$,

$$\frac{\mathrm{d}\mathbf{v}^{[k]}}{\mathrm{d}t} = k|\mathbf{v}|^{k-1}\dot{\mathbf{v}}. \tag{13}$$

Based on the sliding surface eq. (10), the derivative of the sliding surface can be obtained:

$$\dot{\mathbf{s}} = \dot{\mathbf{e}} + \gamma_1\alpha_1 |\mathbf{e}|^{\alpha_1 - 1}\dot{\mathbf{e}} + \gamma_2\alpha_2 |\dot{\mathbf{e}}|^{\alpha_2 - 1}\ddot{\mathbf{e}}. \tag{14}$$

Using eq. (4), $\ddot{\mathbf{e}}$ can be expressed as

$$\begin{aligned} \ddot{\mathbf{e}} &= \ddot{\mathbf{q}} - \ddot{\mathbf{q}}_{\mathbf{d}} \\ &= \mathbf{H}_0(\mathbf{q})^{-1}[\mathbf{u} + \mathbf{D}_{\text{lump}} - \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}_0(\mathbf{q})] - \ddot{\mathbf{q}}_{\mathbf{d}}. \end{aligned} \tag{15}$$

By integrating the definition of $\mathbf{D}_{\text{lum}}$ with Properties 1 and 2, as well as Assumption 1, we can derive the following:

$$\begin{aligned} \|\mathbf{D}_{\text{lum}}\| &\leqslant \bar{d} + h_1\|\ddot{\mathbf{q}}\| + c_1\|\dot{\mathbf{q}}\|^2 + g_1 \\ &\leqslant \bar{d} + h_0h_1(\chi_0 + \bar{d} + g_0g_1) + h_0h_1\chi_1\|\mathbf{q}\| \\ &\quad + h_0h_1(\chi_2 + c_0c_1)\|\dot{\mathbf{q}}\|^2. \end{aligned} \tag{16}$$

Let $\zeta = [\mathbf{e}^{\mathrm{T}}, \dot{\mathbf{e}}^{\mathrm{T}}]^{\mathrm{T}}$, yielding the following inequalities: $\|\mathbf{e}\| \leqslant \|\zeta\|$ and $\|\dot{\mathbf{e}}\| \leqslant \|\zeta\|$. Substituting $\mathbf{q} = \mathbf{e} + \mathbf{q_d}$ and $\dot{\mathbf{q}} = \dot{\mathbf{e}} + \dot{\mathbf{q}}_{\mathbf{d}}$ into the latter inequality, we obtain $\|\mathbf{q}\| \leqslant \|\mathbf{e}\| + \|\mathbf{q_d}\| \leqslant \|\zeta\| + \|\mathbf{q_d}\|$ and $\|\dot{\mathbf{q}}\| \leqslant \|\dot{\mathbf{e}}\| + \|\dot{\mathbf{q}}_{\mathbf{d}}\| \leqslant \|\zeta\| + \|\dot{\mathbf{q}}_{\mathbf{d}}\|$. By combining these inequalities with the boundedness of the input command, inequality eq. (16) can be expressed as

$$\|\mathbf{D}_{\text{lum}}\| \leqslant \mu_0^* + \mu_1^*\|\zeta\| + \mu_2^*\|\zeta\|^2, \tag{17}$$

where

$\mu_0^* \triangleq \bar{d} + h_0h_1[\chi_0 + \bar{d} + g_0g_1 + \chi_1\|\mathbf{q_d}\| + (\chi_2 + c_0c_1)\|\dot{\mathbf{q}}_{\mathbf{d}}\|]$, $\mu_1^* \triangleq h_0h_1[\chi_1 + 2(\chi_2 + c_0c_1)\|\dot{\mathbf{q}}_{\mathbf{d}}\|^2]$, and $\mu_2^* \triangleq h_0h_1(\chi_2 + c_0c_1)$ are unknown but bounded scalars.

Remark 2 From inequality eq. (17), the upper bound of the overall state-dependent uncertainty consists of several terms derived from Properties 1–3. These bounds are system-independent and unknown. The objective is to find a control strategy that enables real-time estimation of these three boundary values (μ_0^*, μ_1^*, μ_2^*) such that the control strategy does not require prior knowledge of the uncertainty boundaries, which is superior to the traditional SMC method.

3.2 Controller design

The control structure framework is illustrated in Figure 2. Considering the system described by eq. (2) with its upper bounding properties and the sliding surface eq. (10) of the nonsingular fast terminal sliding controller, we design the following controller:

$$\mathbf{u} = \mathbf{u}_{\mathbf{eq}} + \mathbf{u}_{\mathbf{re}}, \tag{18}$$

where

$$\begin{aligned} \mathbf{u}_{\mathbf{eq}} &= -\mathbf{H}_0(\mathbf{q})\left[\frac{1}{\gamma_2\alpha_2}\left(\dot{\mathbf{e}}^{[2-\alpha_2]} + \gamma_1\alpha_1 |\mathbf{e}|^{\alpha_1 - 1}\dot{\mathbf{e}}^{[2-\alpha_2]}\right)\right] \\ &\quad + \mathbf{H}_0(\mathbf{q})\ddot{\mathbf{q}}_{\mathbf{d}} + \mathbf{C}_0(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}_0(\mathbf{q}) \end{aligned} \tag{19}$$

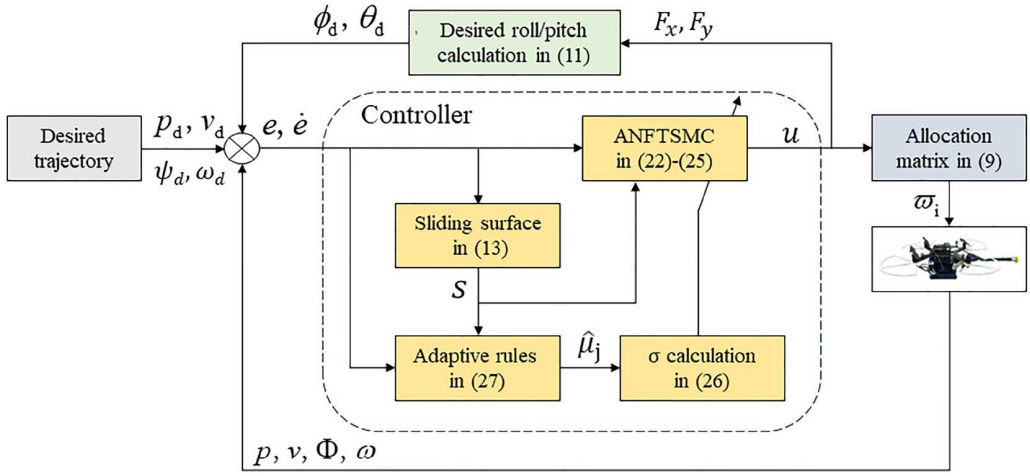

Figure 2 (Color online) Block diagram of control structure framework.

is the equivalent control used to regulate the nominal part of the system [37]; $\dot{\mathbf{e}}^{[2-\alpha_2]} := [\dot{e}_1^{[2-\alpha_2]}, \cdots, \dot{e}_n^{[2-\alpha_2]}]^{\mathrm{T}} \in \mathbb{R}^n$ and $|\mathbf{e}|^{\alpha_1-1} := \mathrm{diag}\left\{|e_1|^{\alpha_1-1}, \cdots, |e_n|^{\alpha_1-1}\right\} \in \mathbb{R}^{n\times n}$. The term

$$\mathbf{u}_{\mathbf{re}} = -\mathbf{H}_0(\mathbf{q})(\mathbf{\Lambda}_0\mathbf{s} + \mathbf{\Lambda}_1\mathrm{sgn}(\mathbf{s})) + \mathbf{u}_{\mathbf{ad}} \tag{20}$$

indicates reaching control, where $\mathbf{\Lambda}_0$ and $\mathbf{\Lambda}_1$ are positive definite diagonal matrices; sgn(·) denotes the standard signum function. For a vector $\mathbf{s}$, $\mathrm{sgn}(\mathbf{s}) = [\mathrm{sgn}(s_1), \cdots, \mathrm{sgn}(s_n)]^{\mathrm{T}} \in \mathbb{R}^n$.

$$\mathbf{u}_{\mathbf{ad}} = -\sigma(t)\mathbf{H}_0(\mathbf{q})\mathrm{sgn}(\mathbf{s}) \tag{21}$$

is the compensatory adaptive control, where $\sigma(t)$ represents a real-time estimation related to the upper bound of uncertainty and can be expressed as follows [19]:

$$\sigma(t) = \hat{\mu}_0(t) + \hat{\mu}_1(t)\|\zeta\| + \hat{\mu}_2(t)\|\zeta\|^2, \tag{22}$$

where the gains $\hat{\mu}_0$, $\hat{\mu}_1$, and $\hat{\mu}_2$ are the real-time estimation of the unknown bounded scalars μ_0^*, μ_1^*, and μ_2^* in eq. (17). These gains are updated online according to the adaptive rules below:

$$\dot{\hat{\mu}}_j(t) = -\delta_j\hat{\mu}_j(t) + \Pi(\dot{e})\|\mathbf{s}\|\|\zeta\|^j, \tag{23}$$

where $\delta_j (j = 0, 1, 2)$ are positive design gains, and the initial value of $\hat{\mu}_j(t) (j = 0, 1, 2)$ should satisfy the condition $\hat{\mu}_j(0) > 0$. $\Pi(\dot{e})$ is defined as

$$\Pi(\dot{e}) := \inf_{1\leqslant i\leqslant n}\left\{\gamma_2\alpha_2\left|\dot{e}_i\right|^{\alpha_2-1}\right\}, \tag{24}$$

where for any matrix, inf{·} indicates an inferior bound.

Integrating the adaptive rule eq. (23) from 0 to t, we get

$$\hat{\mu}_j(t) = \mathrm{e}^{-\delta_j t}\hat{\mu}_j(0) + \int_0^t \mathrm{e}^{-\delta_j(t-\tau)}\Pi(\dot{e})\|\mathbf{s}\|\|\zeta\|^j \mathrm{d}\tau. \tag{25}$$

Since the initial condition $\hat{\mu}_j(0) > 0$, and $\Pi(\dot{e})\|\mathbf{s}\|\|\zeta\|^j \geqslant 0$, we obtain

$$\hat{\mu}_j(t) \geqslant 0, \forall t \geqslant 0. \tag{26}$$

Assumption 2 [33] Given that the upper bound of $\mathbf{D}_{\mathrm{lum}}$ can be expressed as in inequality eq. (17), the following assumption is presented:

$$\frac{\gamma_2\alpha_2}{\Pi(\dot{e})}\left\|\left|\dot{\mathbf{e}}\right|^{\alpha_2-1}\mathbf{H}_0(\mathbf{q})^{-1}\right\|\|\mathbf{D}_{\mathrm{lum}}\| \leqslant \sum_{j=0}^{2}\mu_j\|\zeta\|^j, \tag{27}$$

where $\mu_j (j = 0, 1, 2)$ are unknown positive constants and $\dot{e}_i \neq 0$.

The performance and robustness of the proposed controller are critically determined by selecting the appropriate parameters. The sliding surface parameters γ_1, γ_2, α_1, and α_2 define the dynamics of the sliding surface in eq. (10) and consequently govern the convergence rate both near and far from the equilibrium. Powers α_1 and α_2 must satisfy the nonsingularity condition eq. (11). A larger α_1 accelerates convergence when the error $\mathbf{e}$ is large, whereas a smaller α_2 (closer to 1) enhances the terminal attraction effect when the error derivative $\dot{\mathbf{e}}$ is small, ensuring fast finite-time convergence.

The positive diagonal gains γ_1 and γ_2 weight the contributions of the error and its derivative. Increasing γ_1 places more emphasis on reducing the error $\mathbf{e}$, whereas increasing γ_2 emphasizes the reduction of the error derivative $\dot{\mathbf{e}}$. The values of γ_1 and γ_2 should be tuned to achieve a desired balance between damping and responsiveness, avoiding excessive control effort.

The reaching law parameters in eq. (20) control the dynamics of reaching the sliding surface. The positive definite diagonal matrix $\mathbf{\Lambda}_0$ governs the exponential decay rate of the sliding variable $\mathbf{s}$ once near the sliding surface. A larger $\mathbf{\Lambda}_0$

accelerates the reaching phase but may lead to increased control chattering. The nonlinear term $\mathbf{\Lambda}_1$ provides a constant rate of convergence, whereas a large $\mathbf{\Lambda}_1$ is the primary source of chattering.

The positive gains δ_j ($j = 0, 1, 2$) in the adaptation law eq. (23) play a critical role in guaranteeing the global stability and robustness of the closed-loop system. Larger values of δ_j enhance the robustness of the parameter estimation process by decisively preventing unbounded parameter drift in the presence of disturbances or measurement noise, thus ensuring global stability. Conversely, this robustness mechanism introduces a trade-off whereby larger δ_j values inherently lead to a larger steady-state error in the estimation of the uncertainty bounds, which can marginally reduce the ultimate tracking precision. Therefore, δ_j should be tuned to achieve a balance between robust stability and accurate estimation.

Consequently, the parameters should be sequentially tuned to systematically balance the nominal performance, robustness, and smooth adaptation. First, the sliding surface parameters γ_1, γ_2, α_1, and α_2 are adjusted to achieve satisfactory transient and steady-state performance. Subsequently, the reaching law gains $\mathbf{\Lambda}_0$ and $\mathbf{\Lambda}_1$ are tuned to ensure a swift and robust reaching phase while mitigating chattering. Finally, the adaptive gains δ_j are selected to achieve a suitable compromise between accurate uncertainty estimation and robustness to noise.

Remark 3 [33] In practical application, to mitigate chattering caused by the discontinuous signum function, the continuous function $\frac{\mathbf{s}}{\|\mathbf{s}\|+l_0}$ is adopted to replace the traditional signum function $\mathrm{sgn}(\mathbf{s})$, where l_0 is a small positive constant.

The parameter l_0 directly governs the trade-off between chattering suppression and tracking precision. A larger l_0 results in a smoother control signal and more effective chattering reduction, but introduces a bounded steady-state tracking error, as the system operates within a boundary layer around the sliding surface. Therefore, l_0 is selected to achieve a trade-off between chattering suppression and tracking accuracy.

3.3 Stability analysis

Theorem 1 [33,37] Considering the uncertain quadrotor system described in eq. (2) under lumped disturbances, with the nonlinear sliding surface described in eq. (10) and the controller defined in eq. (18), the system trajectory will quickly converge in finite time from any initial condition, and a singularity will not occur throughout the process.

Proof The proof is divided into three steps. In the first step, we prove that the nonsingular sliding surface $\mathbf{s}$ defined by eq. (10) can converge to the vicinity of $\mathbf{s} = 0$ in finite time. In the second step, we prove that $\dot{e}_i = 0$ is not an attractor, and that no singularity can occur throughout the entire process under the controller defined by (18). In the third step, we prove the finite-time stability of the tracking error $\mathbf{e}$ in the sliding phase.

Step 1: Consider the following Lyapunov function:

$$V_1 = \frac{1}{2}\mathbf{s}^{\mathrm{T}}\mathbf{s} + \sum_{j=0}^{2}\frac{1}{2}\tilde{\mu}_j^{\,2}, \tag{28}$$

where $\tilde{\mu}_j = \mu_j - \hat{\mu}_j(t)$ (for $j = 0, 1, 2$); μ_j and $\hat{\mu}_j(t)$ are defined in eqs. (27) and (22), respectively.

Using eq. (14), the time derivative of V_1 can be expressed as

$$\begin{aligned}\dot{V}_1 &= \mathbf{s}^{\mathrm{T}}\dot{\mathbf{s}} - \sum_{j=0}^{2}\tilde{\mu}_j\dot{\hat{\mu}}_j \\ &= \mathbf{s}^{\mathrm{T}}(\dot{\mathbf{e}} + \gamma_1\alpha_1 \left|\,\mathbf{e}\,\right|^{\alpha_1-1}\dot{\mathbf{e}} + \gamma_2\alpha_2 \left|\,\dot{\mathbf{e}}\,\right|^{\alpha_2-1}\ddot{\mathbf{e}}) \\ &\quad - \sum_{j=0}^{2}\tilde{\mu}_j\dot{\hat{\mu}}_j.\end{aligned} \tag{29}$$

By substituting eq. (15) and the composite control laws eqs. (18) and (19) into eq. (29) and using the adaptive rule eq. (23), we obtain

$$\begin{aligned}\dot{V}_1 &= \mathbf{s}^{\mathrm{T}}\cdot\gamma_2\alpha_2 \left|\,\dot{\mathbf{e}}\,\right|^{\alpha_2-1}\mathbf{H}(\mathbf{q})^{-1}(\mathbf{u}_{\mathbf{re}} + \mathbf{D}_{\mathrm{lum}}) \\ &\quad - \sum_{j=0}^{2}\tilde{\mu}_j(-\delta_j\hat{\mu}_j(t) + \Pi(\dot{e})\|\mathbf{s}\|\|\zeta\|^j).\end{aligned} \tag{30}$$

Using the control laws (20)–(21) and based on eqs. (24) and (26), and Assumption 2, the derivative of V_1 can be expressed as

$$\begin{aligned}\dot{V}_1 &\leqslant -\Pi(\dot{e})\mathbf{s}^{\mathrm{T}}\mathbf{\Lambda}_0\mathbf{s} - \Pi(\dot{e})\mathbf{\Lambda}_1\|\mathbf{s}\| \\ &\quad - \|\mathbf{s}\|\Pi(\dot{e})\sum_{j=0}^{2}\hat{\mu}_j(t)\|\zeta\|^j + \|\mathbf{s}\|\Pi(\dot{e})\sum_{j=0}^{2}\mu_j(t)\|\zeta\|^j \\ &\quad - \sum_{j=0}^{2}\tilde{\mu}_j(-\delta_j\hat{\mu}_j(t) + \Pi(\dot{e})\|\mathbf{s}\|\|\zeta\|^j) \\ &\leqslant -\Pi(\dot{e})\mathbf{\Lambda}_1\|\mathbf{s}\| + \sum_{j=0}^{2}\delta_j\tilde{\mu}_j\hat{\mu}_j(t).\end{aligned} \tag{31}$$

Consider that

$$\begin{aligned}\tilde{\mu}_j\hat{\mu}_j(t) &= \tilde{\mu}_j(\mu_j - \tilde{\mu}_j) = \tilde{\mu}_j\mu_j - \tilde{\mu}_j^{\,2} \\ &= -\left(\frac{\tilde{\mu}_j}{\sqrt{2}} - \frac{\mu_j}{\sqrt{2}}\right)^2 - \frac{\tilde{\mu}_j^{\,2}}{2} + \frac{\mu_j^2}{2} \\ &\leqslant -\frac{|\tilde{\mu}_j|^2}{2} + \frac{|\mu_j|^2}{2},\end{aligned}$$

then

$$\begin{aligned}\dot{V}_1 &\leqslant -\Pi(\dot{e})\mathbf{\Lambda}_1\|\mathbf{s}\| + \sum_{j=0}^{2}\delta_j\left(-\frac{|\tilde{\mu}_j|^2}{2}+\frac{|\mu_j|^2}{2}\right)\\ &= -\Pi(\dot{e})\mathbf{\Lambda}_1\|\mathbf{s}\| - \sum_{j=0}^{2}\frac{\delta_j}{2}\left|\tilde{\mu}_j\right|\\ &\quad+\sum_{j=0}^{2}\frac{\delta_j}{2}\left(\left|\mu_j\right|^2-\left|\tilde{\mu}_j\right|^2+\left|\tilde{\mu}_j\right|-\frac{1}{4}+\frac{1}{4}\right)\\ &\leqslant -\Pi(\dot{e})\mathbf{\Lambda}_1\|\mathbf{s}\| - \sum_{j=0}^{2}\frac{\delta_j}{2}\left|\tilde{\mu}_j\right| + \sum_{j=0}^{2}\frac{\delta_j}{2}\left(\left|\mu_j\right|^2+\frac{1}{4}\right)\\ &\leqslant -\sqrt{2}\min\left\{\Pi(\dot{e})\lambda_{\min}(\mathbf{\Lambda}_1),\frac{1}{2}\min\left\{\delta_j\right\}\right\}V_1^{\frac{1}{2}}+\varpi,\end{aligned} \tag{32}$$

where $\lambda_{\min}(\mathbf{\Lambda}_1)$ denotes the minimum eigenvalue of the matrix $\mathbf{\Lambda}_1$, and $\varpi = \sum_{j=0}^{2}\frac{\delta_j}{2}\left(\left|\mu_j\right|^2+\frac{1}{4}\right)$.

By using Lemma 1, the decreasing Lyapunov function $V_1(x)$ yields: $V_1^{\frac{1}{2}} \leqslant \frac{\varpi}{(1-\varepsilon_0)\sqrt{2}\min\left\{\Pi(\dot{e})\lambda_{\min}(\mathbf{\Lambda}_1),\min\left\{\delta_j\right\}/2\right\}}$, which implies that the system state will enter into the bounded vicinity of $\mathbf{s}=0$ in finite time $T_1 \leqslant \frac{\sqrt{2}(V_1(T_0))^{1/2}}{\varepsilon_0\min\left\{\Pi(\dot{e})\lambda_{\min}(\mathbf{\Lambda}_1),\min\left\{\delta_j\right\}/2\right\}}$, with the bounded region given by $\|\mathbf{s}\| \leqslant \bar{s} = \frac{\varpi}{(1-\varepsilon_0)\min\left\{\Pi(\dot{e})\lambda_{\min}(\mathbf{\Lambda}_1),\min\left\{\delta_j\right\}/2\right\}}$, where $0<\varepsilon_0<1$ and $V_1(T_0)$ denotes the initial value of V_1.

Step 2: For the sliding mode surface eq. (10), there are three cases that lead to $s_i \neq 0$: (1) $e_i \neq 0$, $\dot{e}_i \neq 0$; (2) $e_i = 0$, $\dot{e}_i \neq 0$; (3) $e_i \neq 0$, and $\dot{e}_i = 0$. Both cases (1) and (2) satisfy $\dot{e}_i \neq 0$ such that $\Pi(\dot{e}) > 0$ and $-\sqrt{2}\min\left\{\Pi(\dot{e})\lambda_{\min}(\mathbf{\Lambda}_1),\frac{1}{2}\min\left\{\delta_j\right\}\right\} < 0$ in eq. (32). Thus, the stability condition of Lemma 2 is satisfied when $\dot{e}_i \neq 0$ and the system will rapidly enter the region near the sliding mode surface $\mathbf{s}=0$ in finite time.

Considering case (3), where $e_i \neq 0$ and $\dot{e}_i = 0$, substituting eqs. (19)–(21) into eq. (15) yields

$$\ddot{e}_i = -(\mathbf{\Lambda}_0)_i s_i - [(\mathbf{\Lambda}_1)_i + \sigma(t)]\mathrm{sgn}(s_i) + \Omega_i, \tag{33}$$

where $\Omega = \mathbf{H}(\mathbf{q})^{-1}\mathbf{D}_{\mathbf{lum}}$; $(\mathbf{\Lambda}_0)_i$, $(\mathbf{\Lambda}_1)_i$, and Ω_i represent the ith elements of $\mathbf{\Lambda}_0$, $\mathbf{\Lambda}_1$, and Ω, respectively.

According to Assumption 2, the definition of $\Pi(\dot{e})$ in eq. (24), and $\frac{\gamma_2\alpha_2\left\|\dot{\mathbf{e}}\left|^{\alpha_2-1}\right.\right\|}{\Pi(\dot{e})} \geqslant 1$, we have

$$\|\mathbf{H}_0(\mathbf{q})^{-1}\|\|\mathbf{D}_{\mathrm{lum}}\| \leqslant \sum_{j=0}^{2}\mu_j\|\zeta\|^j. \tag{34}$$

From the proof in step 1, we determine that $\sum_{j=0}^{2}\tilde{\mu}_j$ converges in finite time. Thus, by using eqs. (22) and (34), $(\mathbf{\Lambda}_1)_i$ can be chosen to satisfy $(\mathbf{\Lambda}_1)_i > \sum_{j=0}^{2}\tilde{\mu}_j\|\zeta\|^j$ so that (1) $\ddot{e}_i < 0$ when $s_i > 0$; (2) $\ddot{e}_i > 0$ when $s_i < 0$. This means that $\dot{e}_i = 0$ is not an attractor and $\dot{e}_i = 0$ is not a stable point of the system. Moreover, controller eq. (18) does not contain any negative fractional powers because $\alpha_1 > \alpha_2$ and $1 < \alpha_2 < 2$. Therefore, the sliding mode surface s along eq. (10) can converge to the bounded set in finite time without any singularity.

Step 3: Another Lyapunov function is selected as

$$V_2 = \frac{1}{2}\mathbf{e}^{\mathrm{T}}\mathbf{e}. \tag{35}$$

Taking the derivative of V_2 and substituting $\dot{\mathbf{e}}$ derived from eq. (10) into $\dot{V}_2$, we get

$$\begin{aligned}\dot{V}_2 &= \mathbf{e}^{\mathrm{T}}\dot{\mathbf{e}}\\ &= \mathbf{e}^{\mathrm{T}}\left|\frac{1}{\gamma_2}(\mathbf{s}-\mathbf{e}-\gamma_1\mathbf{e}^{[\alpha_1]})\right|^{\frac{1}{\alpha_2}}\mathrm{sgn}(\mathbf{s}-\mathbf{e}-\gamma_1\mathbf{e}^{[\alpha_1]})\\ &= \sum_{i=1}^{n}e_i\left|\frac{1}{\gamma_2}(s_i-e_i-\gamma_1 e_i^{[\alpha_1]})\right|^{\frac{1}{\alpha_2}}\mathrm{sgn}(s_i-e_i-\gamma_1 e_i^{[\alpha_1]}),\end{aligned} \tag{36}$$

where

$$\left|\frac{1}{\gamma_2}(\mathbf{s}-\mathbf{e}-\gamma_1\mathbf{e}^{[\alpha_1]})\right| = \mathrm{diag}\left\{\left|\frac{1}{\gamma_2}(s_1-e_1-\gamma_1 e_1^{[\alpha_1]})\right|^k,\cdots,\left|\frac{1}{\gamma_2}(s_n-e_n-\gamma_1 e_n^{[\alpha_1]})\right|^k\right\} \in \mathbb{R}^{n\times n}.$$

Note that $\left|s_i\right| > \left|e_i+\gamma_1 e_i^{[\alpha_1]}\right|$ ensures that as $\mathbf{s}$ converges in finite time, $\mathbf{e}$ will also converge. Otherwise, there exists a positive constant ε_1 satisfying $\left|e_i+\gamma_1 e_i^{[\alpha_1]}\right| \geqslant \left|s_i\right| + \varepsilon_1$ so that $\dot{V}_2$ can be expressed as

$$\begin{aligned}\dot{V}_2 &= -\left(\frac{1}{\gamma_2}\right)^{\frac{1}{\alpha_2}}\sum_{i=1}^{n}e_i\left|s_i-e_i-\gamma_1 e_i^{[\alpha_1]}\right|^{\frac{1}{\alpha_2}}\mathrm{sgn}\left(e_i+\gamma_1 e_i^{[\alpha_1]}\right)\\ &\leqslant -\left(\frac{1}{\gamma_2}\right)^{\frac{1}{\alpha_2}}\sum_{i=1}^{n}\left|e_i\right|\left(\left|e_i-\gamma_1 e_i^{[\alpha_1]}\right|^{\frac{1}{\alpha_2}}-\left|s_i\right|^{\frac{1}{\alpha_2}}\right)\\ &\leqslant -\left(\frac{\varepsilon_2}{\gamma_2}\right)^{\frac{1}{\alpha_2}}\sum_{i=1}^{n}\left|e_i\right|,\end{aligned} \tag{37}$$

where ε_2 is a positive constant. By utilizing Lemma 3, we obtain $2V_2 = \mathbf{e}^{\mathrm{T}}\mathbf{e} = \sum_{i=1}^{n}e_i^2 \leqslant \left(\sum_{i=1}^{n}\left|e_i\right|\right)^2$. Thus, the time derivative of V_2 can be written as

$$\dot{V}_2 \leqslant -\sqrt{2}\left(\frac{\varepsilon_2}{\gamma_2}\right)^{\frac{1}{\alpha_2}} V_2^{\frac{1}{2}}. \tag{38}$$

By applying Lemma 1 to eq. (38), $\mathbf{e}$ can converge to a bounded region in finite time $T_2 = T_1 + \sqrt{2}\left(\frac{\gamma_2}{\varepsilon_2}\right)^{\frac{1}{\alpha_2}} V^{\frac{1}{2}}(T_1)$, where $0 < \varepsilon_2 < 1$. Therefore, the proof of Theorem 1 is hereby completed.

4 Numerical simulations

This section demonstrates the effectiveness of the proposed controller relative to that of several other controllers based on comparative simulations.

4.1 Simulation settings

A simulation model of the quadrotor is established in MATLAB. The initial state of the system is stationary, with the quadrotor's mass set to m=4.563 kg, and its inertia tensor defined as I=diag{0.082, 0.0845, 0.1377} kg m^2. The quadrotor is tasked with tracking the target trajectory $\mathbf{p_d} = \left[\sin\left(\frac{\pi}{8}\pi t\right), 0.8\sin\left(\frac{\pi}{4}\pi t\right), 0.4\sin\left(\frac{\pi}{8}\pi t\right)\right]^{\mathrm{T}}$ m while maintaining a yaw angle ψ_d of 0°. At time $t = 0$, the initial position and attitude of the quadrotor are set to $\mathbf{q}(0) = [0.4,\ 0.4,\ 0.2,\ 0,\ 0,\ 0]^{\mathrm{T}}$. To validate the control performance under lumped disturbances, the simulation incorporates two key disturbance components: the modeling inaccuracies are set to 10% of the nominal values of the physical model parameters of the UAM, whereas external disturbances are designed to be state-dependent as $\mathbf{D}_{\mathrm{ext}} = 0.5\cos(0.5t) * \mathbf{1}_{6\times1} + 0.5\mathbf{q}$, with $\mathbf{1}_{6\times1} = [1,1,1,1,1,1]^{\mathrm{T}}$. The selected parameters for the proposed controller are summarized in Table 1. Additionally, the constant l_0, as mentioned in Remark 3, is set to 0.01.

Table 1 Parameters for controllers in simulations

Controllers	Parameters
Proposed	$g_1 = 0.4,\ g_2 = 0.1,\ a_1 = \frac{20}{9},\ a_2 = \frac{4}{3},$ $\mathbf{\Lambda}_0 = \mathrm{diag}\{0.3,\ 0.3,\ 0.6,\ 1,\ 1,\ 1\},$ $\mathbf{\Lambda}_1 = \mathrm{diag}\{2,\ 2,\ 2,\ 2,\ 2,\ 2\},$ $\delta_0 = 0.01,\ \delta_1 = 0.01,\ \delta_2 = 0.01,$ $\hat{\mu}_0(0) = 2,\ \hat{\mu}_1(0) = 2,\ \hat{\mu}_2(0) = 2$
Method 1	$\gamma_1 = 0.4,\ \gamma_2 = 0.1,\ \alpha_1 = \frac{20}{9},\ \alpha_2 = \frac{4}{3}$ $\mathbf{\Lambda}_0 = \mathrm{diag}\{4, 4, 8, 40, 40, 40\},$ $\mathbf{\Lambda}_1 = \mathrm{diag}\{3, 3, 3, 20, 20, 20\}$
Method 2	$\mathbf{\Phi} = \mathrm{diag}\{10,\ 10,\ 20,\ 1,\ 1,\ 1\},$ $\mathbf{\Lambda} = \mathrm{diag}\{14,\ 14,\ 22,\ 2,\ 2,\ 5\},$ $\alpha_0 = 1,\ \alpha_1 = 1,\ \alpha_2 = 1,$ $\widehat{K}_0(0) = 0.1,\ \widehat{K}_1(0) = 0.1,\ \widehat{K}_2(0) = 0.1$
Method 3	$\widehat{\Delta}(0) = \mathbf{0}_{6\times1},\ \mathbf{K}_v = \mathrm{diag}\{10,\ 10,\ 10,\ 8,\ 8,\ 8\},$ $\mathbf{K}_p = \mathrm{diag}\{100,\ 100,\ 200,\ 10,\ 10,\ 10\},$ $\mathbf{K}_\Gamma = \mathrm{diag}\{0.5,\ 1,\ 2,\ 2,\ 2,\ 2\},$ $\mathbf{\Lambda} = \mathrm{diag}\{4,\ 4,\ 4,\ 0.1,\ 0.1,\ 0.1\}$

4.2 Simulation results

To verify the effectiveness of the proposed control method, the quadrotor is set to track the target trajectory. Figure 3 presents a comprehensive evaluation of the tracking performance of the quadrotor. Figure 3(a) and (b) show the tracking performance of the position and attitude, respectively, whereas Figure 3(c) and (d) show the corresponding linear and angular velocities. The convergence of the tracking errors for the position and attitude is displayed in Figure 3(e), and Figure 3(f) shows the evolution of the sliding surfaces. Figure 3(a)–(d) demonstrate that the system rapidly converges to the desired trajectories, confirming effective tracking performance. Moreover, after a brief transient phase caused by initial condition discrepancies, both the position and velocity errors are driven to negligible levels, as evidenced in Figure 3(e). Figure 3(f) shows the time history of the sliding surface variables. All sliding variables converge to a bounded region near zero within approximately 1.5 s, confirming the finite-time reachability of the sliding mode and the robustness of the controller against model uncertainties and external disturbances, which is consistent with the theoretical finite-time stability analysis presented in Section 3.3.

4.3 Simulation comparisons

To verify the effectiveness of the proposed controller, besides a 10% model uncertainty, the system was also exposed to a virtual external disturbance force and torque during the 5th to 10th second intervals. The external perturbation can be expressed as

$$\mathbf{D}_{\mathrm{ext}} = \begin{cases} [8, 9, 10, 2, 3, 4]^{\mathrm{T}}, & 5 \leqslant t \leqslant 10, \\ 0.5\cos(0.5t) * \mathbf{1}_{6\times1} + 0.5\mathbf{q}, & t < 5 \text{ or } t > 10. \end{cases}$$

Note that the proposed controller consists of equivalent control, reaching control, and adaptive-gain switch control, where the reaching control adopts NFTSMC. For comparative evaluation, the proposed controller is benchmarked against three SMC methodologies. Method 1 (constant-gain switching control):

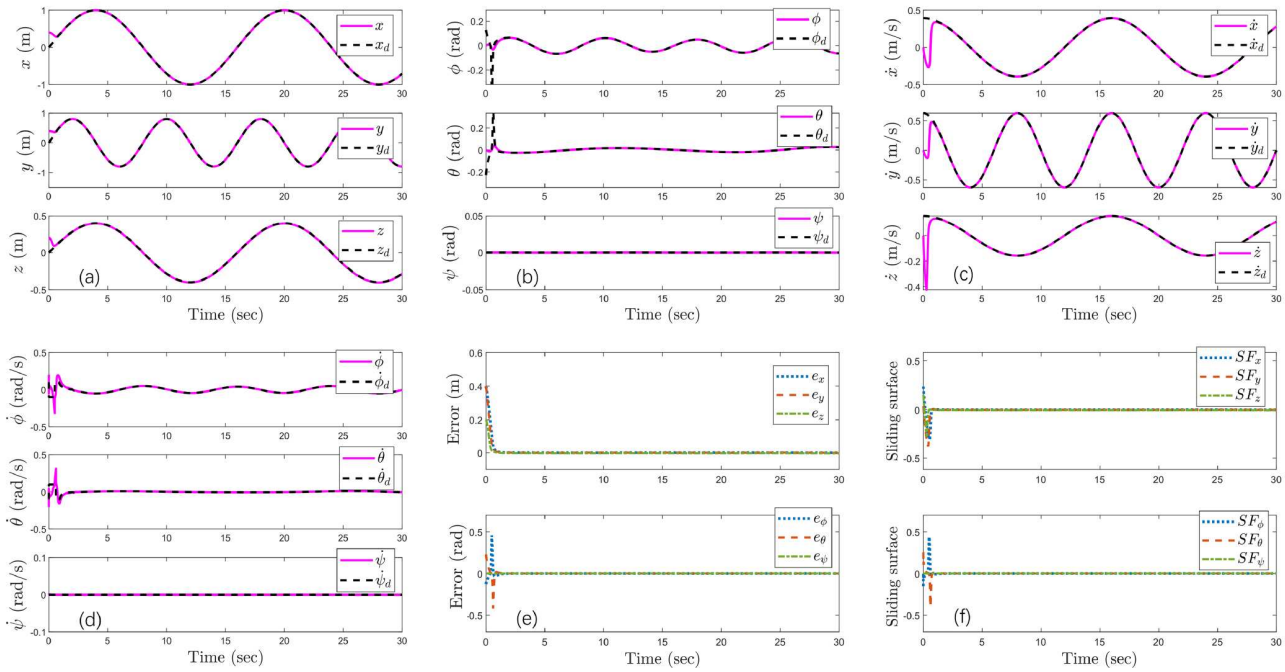

Figure 3 (Color online) Simulation results for tracking performance of the proposed controller. (a) Position tracking performance; (b) attitude tracking performance; (c) velocity tracking performance; (d) angular velocity tracking performance; (e) position and attitude error performance; (f) position and attitude sliding surface performance.

$$\boldsymbol{\tau} = -\mathbf{H}(\mathbf{q})\left[\frac{1}{\gamma_2\alpha_2}\left(\dot{\mathbf{e}}^{[2-\alpha_2]} + \gamma_1\alpha_1 \left|\mathbf{e}\right|^{\alpha_1-1}\dot{\mathbf{e}}^{[2-\alpha_2]}\right)\right] - \mathbf{H}(\mathbf{q})(\boldsymbol{\Lambda}_0\mathbf{s} + \boldsymbol{\Lambda}_1\mathrm{sgn}(\mathbf{s})),$$

where the switching gain ρ remains constant, in contrast with the proposed adaptive strategy. Method 2 [19] combines adaptive switching gains with linear SMC-based reaching control and can be expressed as $\boldsymbol{\tau}(t) = -\boldsymbol{\Lambda}\mathbf{s}(t) - \rho(t)\mathrm{sgn}(\mathbf{s}(t))$, with $\rho(t) = \widehat{K}_0(t) + \widehat{K}_1(t)\|\boldsymbol{\xi}(t)\| + \widehat{K}_2(t)\|\boldsymbol{\xi}(t)\|^2$. Method 3 [31] is an ASMC, where the form of the ASMC is expressed as $\boldsymbol{\tau} = \widehat{\mathbf{M}}\ddot{\mathbf{q}}_r + \widehat{\mathbf{C}}\mathbf{q}_r + \widehat{\mathbf{G}} - \mathbf{K}_v\dot{\mathbf{e}} - \mathbf{K}_p\mathbf{e} + \widehat{\boldsymbol{\Delta}}$, with an adaptive law $\widehat{\boldsymbol{\Delta}} = -\mathbf{K}_I\mathbf{s}$.

For consistent comparison among the simulations, all controllers are evaluated under identical conditions. The system configuration, including the quadrotor dynamics (mass and inertia tensor), initial states, desired trajectory, and disturbance profiles (which incorporate 10% model uncertainty and external disturbances), remains consistent for all methods, as established in the preceding discussion. The parameters for each controller (Table 1) are carefully tuned to achieve favorable overall performance, prioritizing reduction of the steady-state error while comprehensively considering overshoot and the convergence time.

A comparison of the trajectory tracking performance is presented in Figure 4, and quantitative results are provided in Table 2. Figure 4(a) shows the position tracking error performance, Figure 4(b) represents the attitude tracking error performance, and Figure 4(c) shows the position tracking performance from 0 to 2.5 s. Table 2 presents a comparison of the position control performance. As indicated by the maximum overshoot during the 0–4s interval (Figure 4(c) and Table 2), Methods 2 and 3 exhibit significant overshoot and chattering. Furthermore, their mean absolute errors over the 5–12 s and 20–25 s intervals indicate considerable steady-state errors across all axes. In contrast, both Method 1 and the proposed controller, which are both built on the NFTSMC framework, achieve less overshoot and chattering. This confirms the inherent merits of NFTSMC in terms of the convergence speed and stability.

More notably, the proposed scheme achieves faster convergence within 0–4 s and superior disturbance rejection in the 5–12 s period compared with Method 1, demonstrating the effectiveness of the integrated adaptive strategy under external disturbances.

Specifically, the constant switching gain in Method 1 is insufficient to compensate for large external disturbances, resulting in lower adaptability compared to the proposed method. By leveraging the adaptive switching gain, the proposed approach not only demonstrates fast convergence and superior tracking performance under internal model uncertainty but also maintains robust system response under external interference.

5 Experimental studies

5.1 Experimental setup

The developed control scheme is experimentally validated

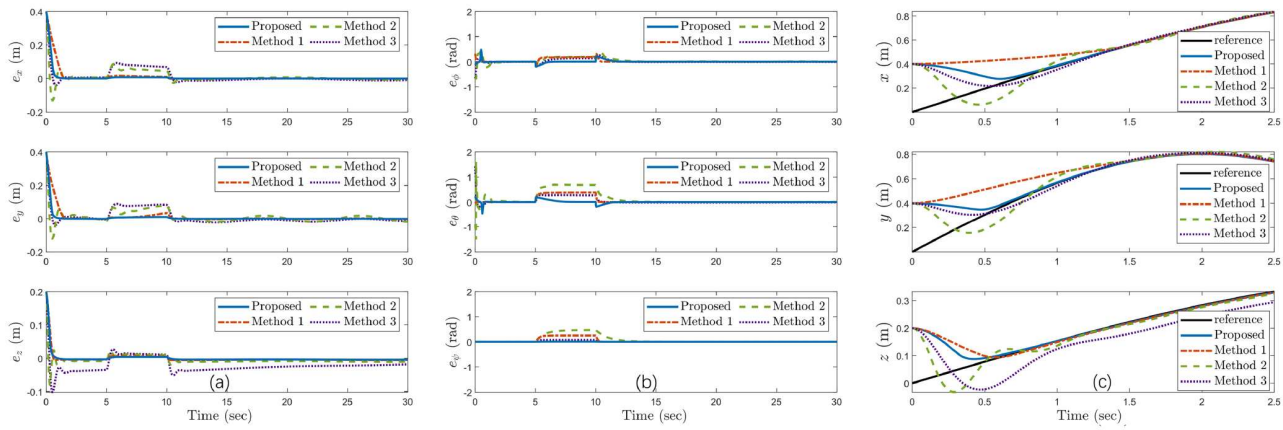

Figure 4 (Color online) Comparison of simulation results for trajectory tracking error performance. (a) Position tracking error performance; (b) attitude tracking error performance; (c) position tracking performance from 0 to 2.5 s.

Table 2 Comparison of position control performance

Method	Dir.	Performance metrics			
		Settling time (s) (0–4 s)	Overshoot Max (m) (0–4 s)	Error Abs Mean (m) (5–12 s)	Error Abs Mean (m) (20–25 s)
Proposed	e_x	1.329	0.0025	0.0055	0.0002
	e_y	1.147	0.001	0.069	0.0003
	e_z	1.173	0.003	0.037	0.0036
Method 1	e_x	1.625	0.002	0.0099	0.0005
	e_y	1.627	0.003	0.0124	0.0015
	e_z	1.296	0.005	0.0059	0.0043
Method 2	e_x	1.887	0.1245	0.0413	0.004
	e_y	1.715	0.1287	0.0436	0.0106
	e_z	1.285	0.0779	0.011	0.0092
Method 3	e_x	1.528	0.0356	0.0587	0.0029
	e_y	1.651	0.0429	0.0609	0.0137
	e_z	1.524	0.0937	0.0201	0.0229

using the setup illustrated in Figure 5. The experimental system comprises a quadrotor platform, an OptiTrack motion capture system with a positioning accuracy of 0.2 mm and a data transmission rate of 100 Hz, a base station for processing and broadcasting motion data, and a ground station for real-time monitoring. The quadrotor platform is equipped with an onboard computer operating at a control frequency of 200 Hz, a flight controller, four brushless motors, a 4-in-1 electronic speed controller (ESC), an SBT674-50 single-axis force sensor with a measurement accuracy of 0.1%, a transducer, and a carbon fiber body frame. The flight controller fuses data from onboard sensors (including gyroscopes, accelerometers, magnetometers, and barometers) and the OptiTrack system to estimate the position, orientation, angular velocity, and linear velocity of the vehicle. The onboard computer executes the control algorithm and sends command signals to the flight controller, which drives the motors via the ESC to adjust the attitude and position of the vehicle. For tasks requiring physical interaction tasks, a carbon fiber manipulator arm with a length of 0.51 m is mounted along the x-axis of the quadrotor body via a 3D-printed connector, and the force sensor is attached to its end effector. The system is powered by a 4S lithium battery, and the total mass of the platform is 2.344 kg.

5.2 Stable flight under wind disturbances

When performing tasks outdoors at high altitudes, a quadrotor is often disturbed by wind and deviates from the expected position or attitude. Resisting interference and quickly restoring its pose are crucial for the quadrotor to complete tasks safely and quickly. In this experiment, an industrial fan is used to produce controlled wind disturbances at 5.8 m/s, simulating realistic outdoor flight

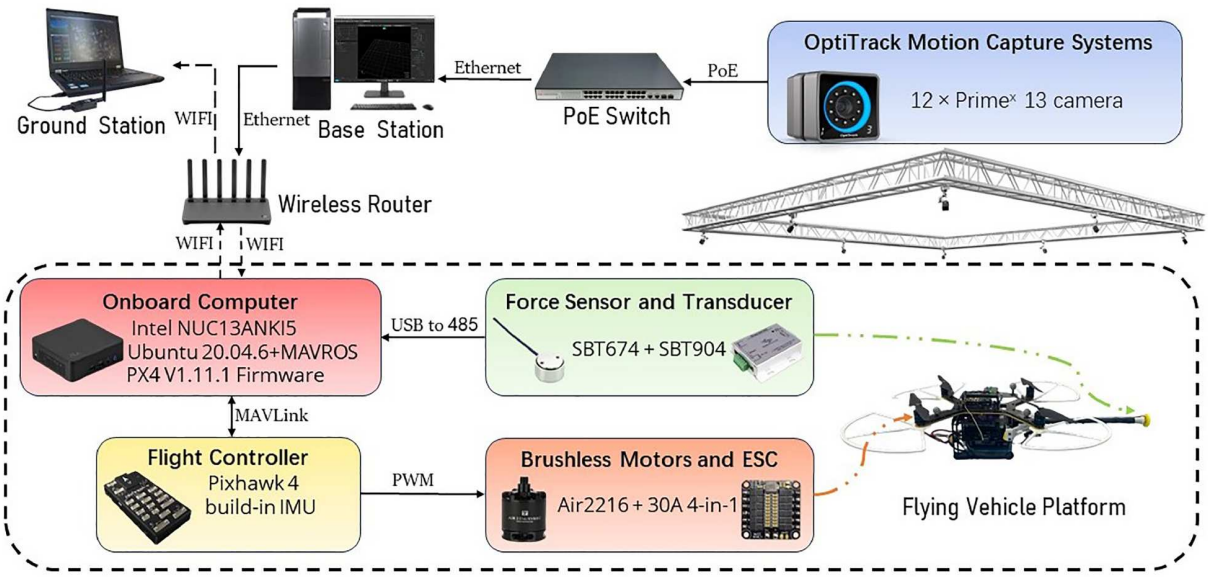

Figure 5 (Color online) Structure of the experimental system.

conditions. We verify and compare the developed controller with the controllers previously introduced in the simulation study, focusing on the capability of the flight platform for rejecting wind disturbances and regaining stability while tracking a desired trajectory.

The quadrotor is initially set to hover near the point of $[0.6,\ -0.7,\ 0.9]^{\mathrm{T}}$ before being tasked to track a lemniscate reference trajectory under the disturbance of an industrial fan (as shown in Figure 6) while maintaining a yaw angle of 0° throughout the process. The desired position trajectory of the UAM follows the law:

$$\mathbf{p_d} = \left[0.8\sin\left(\frac{\pi}{8}\pi t\right), 0.6\sin\left(\frac{\pi}{4}\pi t\right), 0.4\sin\left(\frac{\pi}{8}\pi t\right)\right]^{\mathrm{T}}\mathrm{m},$$

and the initial displacement deviations in the x-, y-, and z-directions are 0.2, 0.3, and 0.15 m, respectively. The parameters for each controller are initially adjusted and selected via a real-time tuning interface on the Gazebo simulation platform, prioritizing reduction of the steady-state error while comprehensively considering the overshoot and convergence time. In real experimental scenarios, further fine-tuning is conducted based on the same set of comprehensive performance criteria.

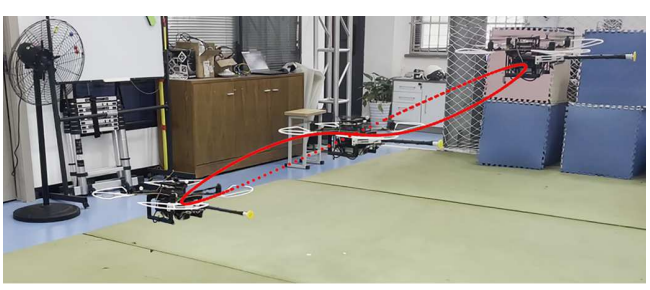

Figure 6 (Color online) Quadrotor tracking a lemniscate reference path (red, dashed line) under the interference of an industrial fan.

Using the parameters listed in Table 3, the tracking performance of the lemniscate reference trajectory is evaluated under four different controllers. Figure 7(a)–(c) present the tracking error performance in the x, y, and z directions, respectively. Under the same conditions, the proposed approach demonstrates the smallest steady-state errors, the fastest recovery to equilibrium, and better anti-disturbance capability. Figure 7(d) illustrates the control thrust input, showing that the proposed method achieves reduced sig-

Table 3 Experimental parameters for controllers

Controllers	Parameters
Proposed	$\gamma_1 = 0.8,\ \gamma_2 = 0.4,\ \alpha_1 = 1.5,\ \alpha_2 = 1.4,$ $\mathbf{\Lambda}_0 = \mathrm{diag}\{2,\ 2,\ 4,\ 12,\ 12,\ 12\},$ $\mathbf{\Lambda}_1 = \mathrm{diag}\{0.1,\ 0.1,\ 0.1,\ 0.8,\ 0.8,\ 0.8\},$ $\delta_0 = 3.6,\ \delta_1 = 3.6,\ \delta_2 = 7.2,$ $\bar{\mu}_0(0) = 0.5,\ \bar{\mu}_1(0) = 0.5,\ \bar{\mu}_2(0) = 0.5$
Method 1	$\gamma_1 = 0.8,\ \gamma_2 = 0.4,\ \alpha_1 = 1.5,\ \alpha_2 = 1.4,$ $\rho = 0.5,\ \mathbf{\Lambda}_0 = \mathrm{diag}\{2,\ 2,\ 4,\ 12,\ 12,\ 12\},$ $\mathbf{\Lambda_1} = \mathrm{diag}\{0.1,\ 0.1,\ 0.1,\ 0.8,\ 0.8,\ 0.8\}$
Method 2	$\mathbf{\Phi} = \mathrm{diag}\{1,\ 1,\ 1,\ 0.2,\ 0.2,\ 0.2\},$ $\mathbf{\Lambda} = \mathrm{diag}\{4,\ 4,\ 4,\ 0.5,\ 0.5,\ 0.5\},$ $\alpha_0 = 3.6,\ \alpha_1 = 3.6,\ \alpha_2 = 7.2,$ $\hat{K}_0(0) = 0.5,\ \hat{K}_1(0) = 0.5,\ \hat{K}_2(0) = 0.5$
Method 3	$\hat{\Delta}(0) = \mathbf{0}_{6\times 1},\ \mathbf{K}_v = \mathrm{diag}\{8,\ 8,\ 8,\ 5,\ 5,\ 5\},$ $\mathbf{K}_p = \mathrm{diag}\{16,\ 16,\ 16,\ 1,\ 1,\ 1\},$ $\mathbf{K}_\Gamma = \mathrm{diag}\{0.1,\ 0.1,\ 0.1,\ 0.2,\ 0.2,\ 0.2\},$ $\mathbf{\Lambda} = \mathrm{diag}\{0.5,\ 0.5,\ 0.5,\ 0.02,\ 0.02,\ 0.02\}$

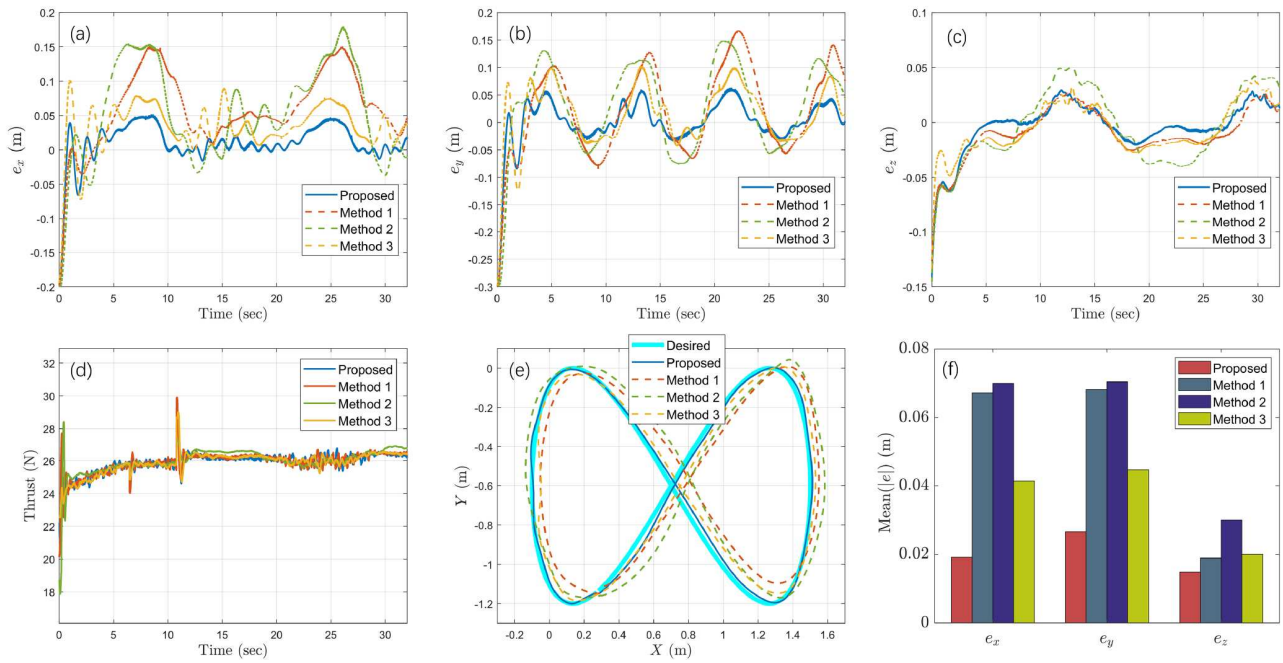

Figure 7 (Color online) Experimental comparison of the ability to track the lemniscate reference path. (a) x-direction position tracking error performance; (b) y-direction position tracking error performance; (c) z-direction position tracking error performance; (d) thrust input; (e) tracking performance in the x-y plane; (f) mean absolute error of tracking performance.

nificant chattering compared with the other methods. Figure 7(e) shows a clear deviation of the quadrotor from the desired trajectory in the positive x-direction under all controllers, owing to the wind disturbance generated by the industrial fan, which primarily blows along the x-axis. Although all trajectories are influenced by the wind, the proposed controller achieves a trajectory closest to the desired path, demonstrating its effectiveness in maintaining its performance under disturbances. This superior performance can be attributed to the adaptive mechanism of the proposed ANFTSMC. The adaptive laws continuously compensate for the wind-induced disturbances, which are state-dependent and time-varying. In contrast, the fixed switching gain employed in Method 1 is insufficient to compensate optimally for the varying disturbance, leading to larger steady-state errors. Among the compared methods, the proposed system has the lowest mean absolute error in the position along the x, y, and z directions during the tracking process, as illustrated in Figure 7(f), further confirming the superior control capability of the proposed method.

5.3 Force control for contact-based inspection of wind turbine blades

The previously described quadrotor, equipped with a force sensor, is used for a maintenance and inspection task involving wind turbine blades. In the laboratory, we set up a scenario in which the manipulator of the quadrotor maintains stable contact with a segment of a wind turbine blade along the x-direction, as illustrated in Figure 8. To fulfill the inspection requirements, the objective is to achieve a stable contact force between the manipulator and the blade. This is accomplished using the impedance control strategy described by eq. (39), a technique validated by Hogan [38] and other researchers:

$$m\ddot{x}_c + b\dot{x}_c + k(x_c - x_r) = f_r - f_m, \tag{39}$$

where x_c and x_r specify the commanded and reference trajectories, f_r and f_m indicate the desired and measured interaction forces, and m, b, and k correspond, respectively, to the virtual mass, damping coefficient, and stiffness of the system, which are the three configurable parameters defining the impedance characteristics.

The contact surface of the electric fan blade is oriented

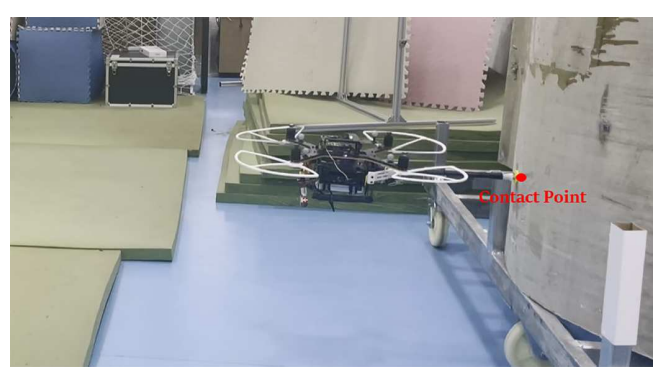

Figure 8 (Color online) Manipulator maintaining stable contact with a segment of a wind turbine blade.

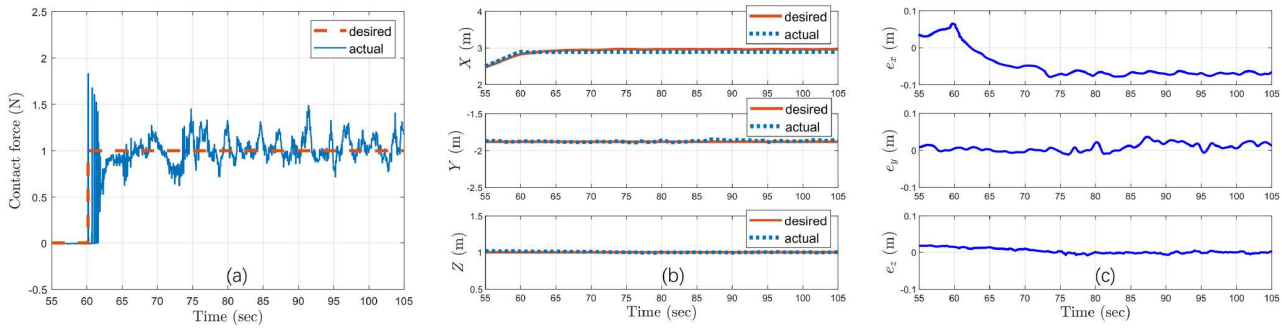

Figure 9 (Color online) Experimental results under impedance controller. (a) Force tracking performance under impedance controller; (b) position tracking performance under impedance controller; (c) position tracking error performance under impedance controller.

perpendicular to the x-axis and positioned at $x_e = 2.88$ m. The quadrotor UAV is directed to approach the surface of the blade with its manipulator arm facing forward. The resulting control performance is shown in Figure 9, using a reference position of , virtual mass , damping coefficient , and stiffness, while maintaining the original position controller parameters and targeting a contact force of 1 N. As shown in Figure 9(a), contact between the quadrotor UAV and the wind turbine blade is established at the 60 s mark. Using the proposed control strategy, the contact force is consistently maintained around the ideal value of 1 N, with the error confined to a limited range. As illustrated in Figure 9(c), the position of the quadrotor along the x-axis showed a steady-state deviation compared with the reference trajectory. This positional offset stems from the contact force, which is inherently reciprocal, thereby establishing the steady-state contact force. Additionally, Figure 9(b) and (c) demonstrate that the position controller exhibits favorable tracking performance in the y- and z-directions.

It is important to note that within this control framework, the commanded position trajectory x_c generated by the impedance controller serves as the control target for the proposed position controller. The impedance control approach essentially transforms the force control issue into a position tracking task, emphasizing the critical role of position tracking performance in ensuring stable and accurate force control. The successful regulation of the contact force in this experimental scenario further validates the effectiveness of the proposed position control method.

6 Conclusion

This study presented an adaptive, nonsingular, fast terminal SMC (ANFTSMC) strategy for a quadrotor UAV, effectively addressing trajectory tracking challenges in the presence of model uncertainties and external disturbances. The proposed control law integrates the NFTSMC concept, ensuring finite-time convergence and avoiding singularities. By employing an adaptive parameter-tuning mechanism, the strategy eliminates the need for prior knowledge of uncertainty bounds and reduces chattering effects. The superiority of the control strategy was validated through extensive simulations and experimental comparisons under wind disturbance and was demonstrated in a force control application during contact with a wind turbine blade. The results reveal that the control method effectively achieves rapid convergence, minimizes steady-state error, and exhibits strong robustness, highlighting its advanced performance and practical applicability in real-world scenarios. Based on the present work, future research will focus primarily on the following two aspects. First, the explicit integration of actuator saturation constraints into the control design will be pursued to enhance the robustness when operating at physical limits. Second, the framework will be extended to manage the full dynamics of aerial manipulators for more complex physical interaction tasks beyond simplified impedance control.

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